Living Village: A Generative Multi-Agent Ecosystem

Assignment 1: Group Formation, Pillars, and Conceptualization

1. Group Roster - *Group JJ*

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2. Selected Pillars

Pillar	How the project uses it
NPC AI / Behavior Trees	Needs-driven behavior trees for sleeping, eating, working, socializing and reacting to emergencies, with relationships and episodic memory.
Pathfinding & Navigation	Recast/Detour NavMesh (recast-navigation-js) with dynamic obstacles, so villagers reroute around new objects and away from danger.
Swarm & Boids Modeling	A flock of 20 to 40 birds using separation, alignment and cohesion, reacting to disturbances such as fire.

3. Project Concept Summary

Living Village is a small, polished 3D village that runs on its own. The user watches it, inspects it and disrupts it.

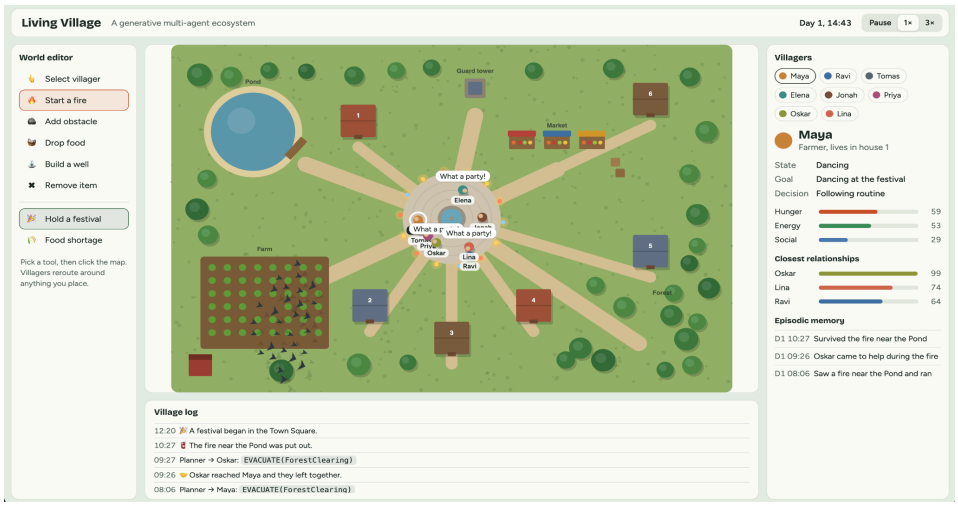
1. **The World & Agents:** Eight unique villagers (each with specific occupations, needs, routines, relationships and episodic memories) and a 30-bird flock inhabit a town featuring houses, a farm, a market and a forest.
2. **Systems Integration:** Villager daily lives are driven by behavior trees and a Recast/Detour NavMesh for dynamic pathfinding. Birds utilize classic Boids rules to flock and scatter.
3. **Interactive Disruptions:** Users can alter the world by starting fires, dropping obstacles, or holding festivals. Agents react logically in real-time guarding the town, fleeing danger, or helping friends and form long-term memories of the events.
4. **Transparent UI:** Clicking any villager opens an inspector detailing their needs, current goals, relationships, and reasoning for their latest actions.
5. **Eventual AI Layer:** An eventual LLM integration will handle complex scenarios by issuing high-level commands (e.g., EVACUATE(TownSquare)), utilizing PostgreSQL with pgvector for memory retrieval. The simulation remains fully functional without it.
6. **Tech Stack:** Next.js, React Three Fiber, and Three.js (Frontend); FastAPI (Backend); connected via REST and WebSockets.

4. AI-Generated Mockups

A simplified 2D top-down version of the planned 3D village.



Fire event. A fire near the pond: villagers evacuate on paths that avoid the flames, the guard runs to put it out, and the birds scatter. The inspector shows Maya's goal and her EVACUATE(ForestClearing) decision.



Festival and memory. Villagers gather to dance in the square. Maya's memory now records the fire and that her friend Oskar came back to help her, which raised their relationship.

5. Interactive Mockup

Link: https://ishashah2303.github.io/living-village/

A vanilla HTML/CSS/JS page demonstrating the core loop using stand-in systems (grid A* for NavMesh, priority selectors for behavior trees, 2D Boids and rule-based planning). Users can trigger events (fires, festivals, shortages), place objects, adjust speed and inspect villagers.

Actual project Simulation Pipeline and Plan:

World Event → NPC Perception → Behavior & Memory → High-Level Decision → Navigation → Animation → Visible Reaction.